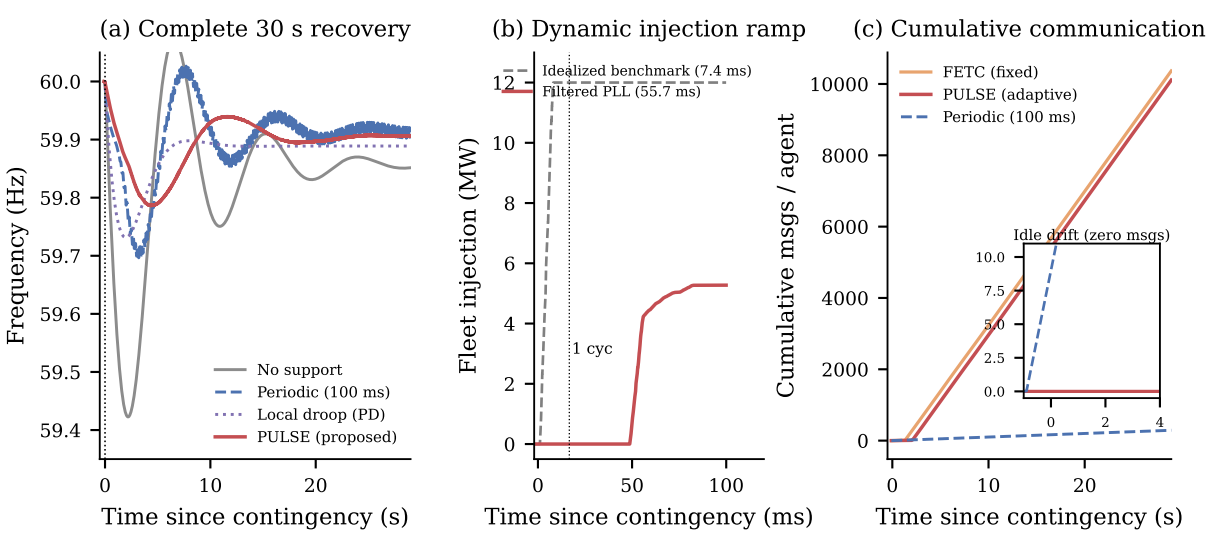
(a) Complete 30 s recovery
Frequency (Hz)
60.0
59.9
59.8
59.7
59.6
59.5
59.4
No support
Periodic (100 ms)
Local droop (PD)
PULSE (proposed)
0
10
20
Time since contingency (s)
(b) Dynamic injection ramp
Idealized benchmark (7.4 ms)
Filtered PLL (55.7 ms)
Fleet injection (MW)
12
10
8
6
4
2
0
1 cyc
0
50
100
Time since contingency (ms)
(c) Cumulative communication
FETC (fixed)
PULSE (adaptive)
Periodic (100 ms)
Cumulative msgs / agent
10000
8000
6000
4000
2000
0
Idle drift (zero msgs)
10.0
7.5
5.0
2.5
0.0
0
2
4
0
10
20
Time since contingency (s)